

SAMSUNG AM DNCDKG SERIES INSTALLATION MANUAL



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Air conditioner

Installation manual

AM*DNCDKG**

- Thank you for purchasing this Samsung air conditioner.
- Before operating this unit, please read this manual carefully and retain it for future reference.

SAMSUNG

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Safety Information

⚠ WARNING

- Hazards or unsafe practices that may result in severe personal injury or death.

⚠ CAUTION

- Hazards or unsafe practices that may result in minor personal injury or property damage.
- Carefully follow the precautions listed below because they are essential to guarantee the safety of the equipment.

⚠ WARNING

- Always disconnect the air conditioner from the power supply before servicing it or accessing its internal components.
- Verify that installation and testing operations are performed by qualified personnel.
- Verify that the air conditioner is not installed in an easily accessible area.

⚠ WARNING

- Carefully read the content of this manual before installing the air conditioner and store the manual in a safe place in order to be able to use it as reference after installation.
- For maximum safety, installers should always carefully read the following warnings.
- Store the operation and installation manual in a safe location and remember to hand it over to the new owner if the air conditioner is sold or transferred.
- This manual explains how to install an indoor unit with a split system with two SAMSUNG units. The use of other types of units with different control systems may damage the units and invalidate the warranty. The manufacturer shall not be responsible for damages arising from the use of non compliant units.
- The manufacturer shall not be responsible for damage originating from unauthorized changes or the improper connection of electric and requirements set forth in the "Operating limits" table, included in the manual, shall immediately invalidate the warranty.
- The air conditioner should be used only for the applications for which it has been designed: the indoor unit is not suitable to be installed in areas used for laundry.
- Do not use the units if damaged. If problems occur, switch the unit off and disconnect it from the power supply.
- In order to prevent electric shocks, fires or injuries, always stop the unit, disable the protection switch and contact SAMSUNG's technical support if the unit produces smoke, if the power cable is hot or damaged or if the unit is very noisy.
- Always remember to inspect the unit, electric connections, refrigerant tubes and protections regularly. These operations should be performed by qualified personnel only.
- The unit contains moving parts, which should always be kept out of the reach of children.
- Do not attempt to repair, move, alter or reinstall the unit. If performed by unauthorized personnel, these operations may cause electric shocks or fires.
- Do not place containers with liquids or other objects on the unit.
- All the materials used for the manufacture and packaging of the air conditioner are recyclable.
- The packing material and exhaust batteries of the remote controller(optional) must be disposed of in accordance with current laws.
- The air conditioner contains a refrigerant that has to be disposed of as special waste. At the end of its life cycle, the air conditioner must be disposed of in authorised centres or returned to the retailer so that it can be disposed of correctly and safely.

Symbol	Meaning
	Flammable gas
	Flammable materials
	Refrigerant safety group
	Read operating manual
	Refer to operating manual
	Read service manual

Safety Information

Installing the unit

WARNING

IMPORTANT: When installing the unit, always remember to connect first the refrigerant tubes, then the electrical lines.

- Always disassemble the electric lines before the refrigerant tubes.
- Upon receipt, inspect the product to verify that it has not been damaged during transport. If the product appears damaged, DO NOT INSTALL it and immediately report the damage to the carrier or retailer (if the installer or the authorized technician has collected the material from the retailer.)
- After completing the installation, always carry out a functional test and provide the instructions on how to operate the air conditioner to the user.
- Do not use the air conditioner in environments with hazardous substances or close to equipment that release free flames to avoid the occurrence of fires, explosions or injuries.
- Our units should be installed in compliance with the spaces shown in the installation manual, to ensure accessibility from both sides and allow repairs or maintenance operations to be carried out. The unit's components should be accessible and easy to disassemble without endangering people and objects. For this reason, when provisions of the installation manual are not complied with, the cost required to access and repair the units (in SAFETY CONDITIONS, as set out in prevailing regulations) with harnesses, ladders, scaffolding or any other elevation system will NOT be considered part of the warranty and will be charged to the end customer.

Power supply line, fuse or circuit breaker

WARNING

- Always make sure that the power supply is compliant with current safety standards. Always install the air conditioner in compliance with current local safety standards.
- Always verify that a suitable grounding connection is available.

- Verify that the voltage and frequency of the power supply comply with the specifications and that the installed power is sufficient to ensure the operation of any other domestic appliance connected to the same electric lines.
- Always verify that the cut-off and protection switches are suitably dimensioned.
- Verify that the air conditioner is connected to the power supply in accordance with the instructions provided in the wiring diagram included in the manual.
- Always verify that electric connections (cable entry, section of leads, protections...) are compliant with the electric specifications and with the instructions provided in the wiring scheme. Always verify that all connections comply with the standards applicable to the installation of air conditioners.
- Devices disconnected from the power supply should be completely disconnected in the condition of overvoltage category.
- Be sure not to perform power cable modification, extension wiring, and multiple wire connection.
 - It may cause electric shock or fire due to poor connection, poor insulation, or current limit override.
 - When extension wiring is required due to power line damage, refer to Step 13 Optional: Extending the power cable in the installation manual.

CAUTION

Make sure that you earth the cables.

- Do not connect the earth wire to the gas pipe, water pipe, lighting rod or telephone wire. If earthing is not complete, electric shock or fire may occur.

Install the circuit breaker.

- If the circuit breaker is not installed, electric shock or fire may occur.

Make sure that the condensed water dripping from the drain hose runs out properly and safely.

Install the power cable and communication cable of the indoor and outdoor unit at least 1m away from the electric appliance.

Install the indoor unit away from lighting apparatus using the ballast.

- If you use the wireless remote control, reception error may occur due to the ballast of the lighting apparatus.

This unit is equipped with electrically powered safety measure. For the safety measures to be effective, the unit must be electrically powered at all times after installation, other than when servicing.

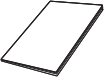
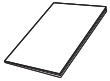
This unit is equipped with a leak detection system for safety. For leak detection to be effective, the unit must be electrically powered at all times after installation, other than when servicing.

Installation Procedure

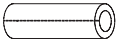
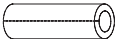
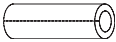
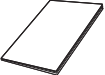
Step 1 Checking and preparing accessories

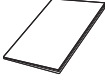
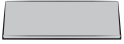
The following accessories are supplied with the indoor unit. The type and quantity may differ depending on the specifications.

Ceiling type (AM056DNC**G***)

Pattern sheet	User's manual
	
Installation manual	Plate Hanger
	
Cable-Tie	
	

Big Ceiling type (AM071/112/140DNC**G***)

Pattern sheet	Insulation cover pipe A
	
Insulation cover pipe B	Insulation drain
	
Flexible hose clamp	Flexible hose
	
Cable-tie	User manual
	

Install manual	Insulation
	
Rubber	
	

Step 2 Choosing the installation location

- Select a convenient location that permits the air to reach every corner of the area to be cooled.
- Pre-plan for easy and short routing of the refrigerant tubing and wiring to the outdoor unit.
- There should be no flammable gas, alkaline, substances present in the air.
- Avoid location where obstacles preventing good air circulation are present.
- Noise prevention should be considered in determining the unit's location.
- The structure, where the unit is to be installed should be strong enough to support the weight of the unit.
- Rigid wall without vibration.
- Where it is not exposed to direct sunshine.
- Where the air filter can be removed and cleaned easily.
- A location where animals cannot access and urinate on the product. Ammonia may be generated.
- The amount of refrigerant to add differs, depending on the installation conditions (e.g., the total piping length and the indoor unit combination), and the minimum indoor-unit installation area depends on the final amount of refrigerant.
Be sure to install by referring to the minimum indoor-unit installation area depending on the final refrigerant amount, provided in the refrigerant amount table in the outdoor unit installation manual.

WARNING

- Because your air conditioner contains R-32 refrigerant, make sure that it is installed, operated, and stored in a room whose floor area is larger than the minimum required floor area specified.

Installation Procedure

- IMPORTANT: It's mandatory to either Follow the table in the Outdoor Unit Installation Manual or follow the federal, state, and/or local regulations regarding the minimum room area allowed for the total refrigerant charge in the system.
- Minimum installation height of indoor unit is 1.8 m for wall, 2.5 m for ceiling, 2.2 m for Ducted.
- R-32-exclusive wired remote control: Equipped with a safety alarm.
- Outdoor unit (AM***DXMD*G/EU): Equipped with a shut-off valve.

Do not install the air conditioner in following places.

- Place where there is mineral oil or arsenic acid. Resin parts flame and the accessories may drop or water may leak. The capacity of the heat exchanger may reduce or the air conditioner may be out of order.
- A place with exposure to mineral oil, oil vapour or cooking area where there is spray (If oil adheres to the heat exchanger, performance degradation, spray or condensation scattering may occur. If oil adheres to a plastic component, the component may deform or get damaged. Such issues may result in a system failure or refrigerant leak.)

- A place with aromatic diffusers, aromatherapy, scented candles or perfumes as the chemicals may react to the product's materials and may result in system failure or refrigerant leaks.
- The place where corrosive gas such as sulphuric acid gas generates from the vent pipe or air outlet.
- The copper pipe or connection pipe may corrode and refrigerant may leak.
- The place where there is a machine that generates electromagnetic waves. The air conditioner may not operate normally due to control system.
- The place where there is a danger of existing combustible gas, carbon fibre or flammable dust.
- The place where thinner or gasoline is handled.
- Gas may leak and it may cause fire.
- The place where is close to heat sources.
- Do not use the indoor unit for preservation of food items, plants, equipment, and art works. This may cause deterioration of their quality.
- Do not install the indoor unit if it has any drainage problem.

Installation conditions for indoor units and wired remote controls

- Make sure to install a dedicated R-32-exclusive wired remote control for each indoor unit.
See the installation examples below for reference.

Make sure to use R-32-exclusive wired remote controls. The product will not operate if an R-32-exclusive wired remote control is not located in the vicinity or if users try to control the product using a common wired remote control.

※ E694: This error occurs if an installed R-32 indoor unit and R-32-exclusive wired remote control are not a correct combination.

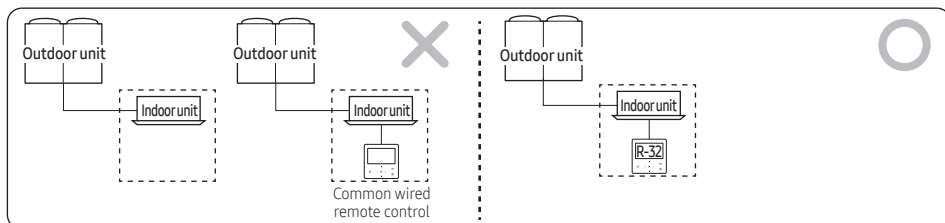
Use R-32-exclusive wired remote controls.

※ MWR-WG01*N

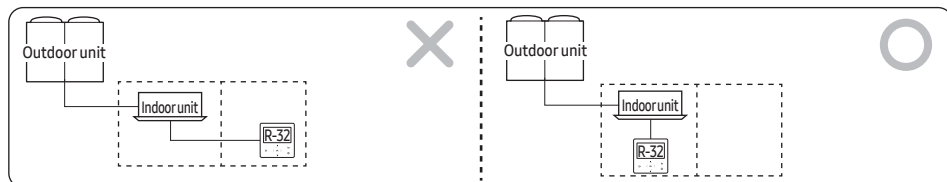
※ R-32-exclusive wired remote controls should be purchased separately.

⚠ WARNING

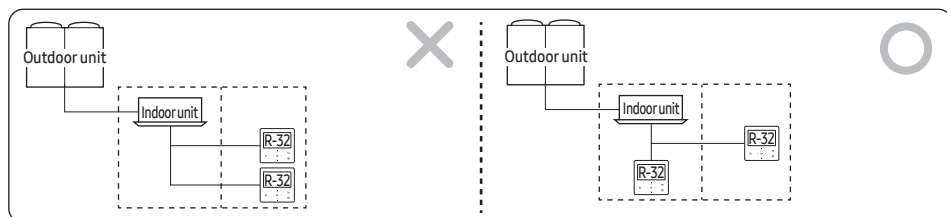
- Samsung Electronics is not responsible for any loss or damage to the product resulting from using anything but the specified wired remote control.



Make sure R-32-exclusive wired remote controls are located in the same room as the indoor units.

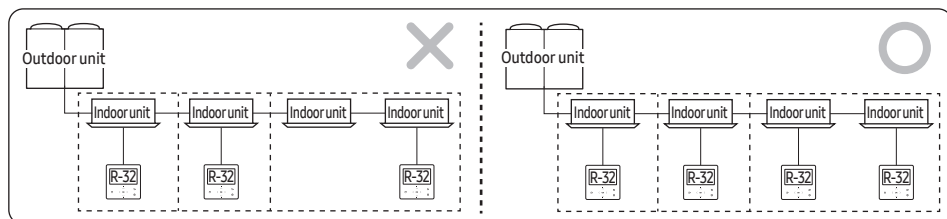


If using two or more R-32-exclusive wired remote controls, at least one of them must be placed in the same room as the indoor units.



Make sure to connect all indoor units to respective R-32-exclusive wired remote controls.

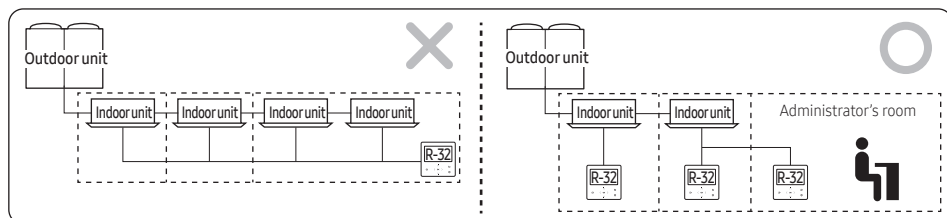
At least one remote control must be installed for each indoor unit, even if multiple indoor units are installed in the same room. Grouped control is not possible.



For the occupancy listed below, the safety alarm system shall also warn at a supervised location, such as the night porter's location, as well as the occupied space:

- rooms, parts of buildings, building where sleeping facilities are provided,
- rooms, parts of buildings, building where people are restricted in their movement,
- rooms, parts of buildings, building where an uncontrolled number of people are present, or
- rooms, parts of buildings, building to which any person has access without being personally acquainted with the necessary safety precautions.

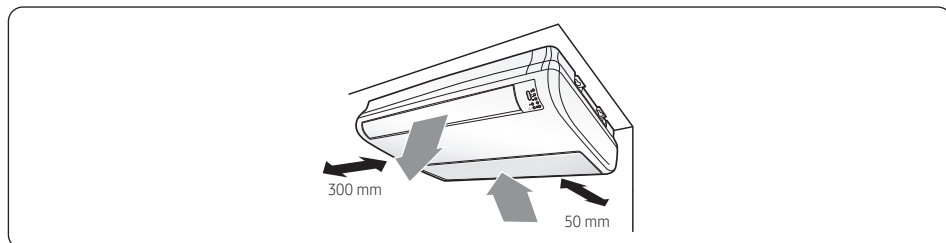
A wired remote control must be installed in the administrator's room, using wired remote control supervisor mode. For details on how to set wired remote control supervisor mode, refer to the wired remote control installation manual.



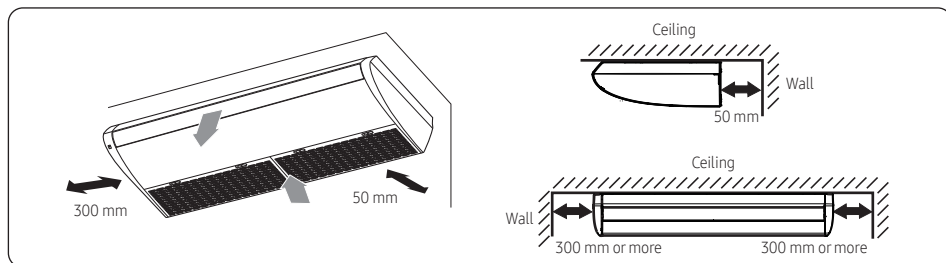
Installation Procedure

Ceiling installation

Ceiling type (AM056DNC**G***)



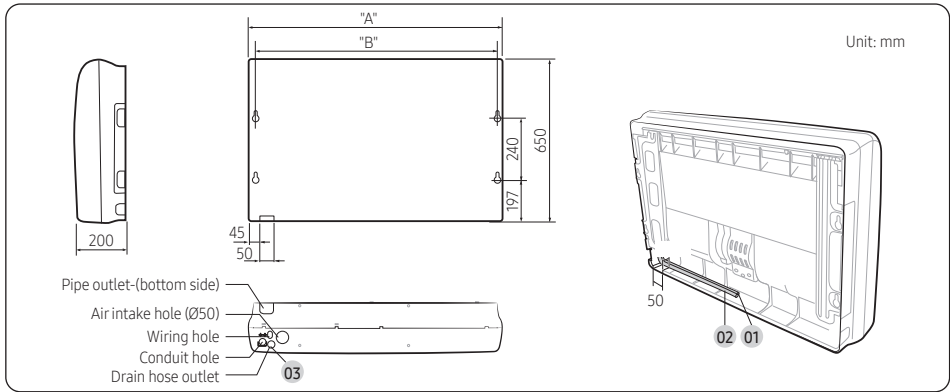
Big Ceiling type (AM071/112/140DNC**G***)



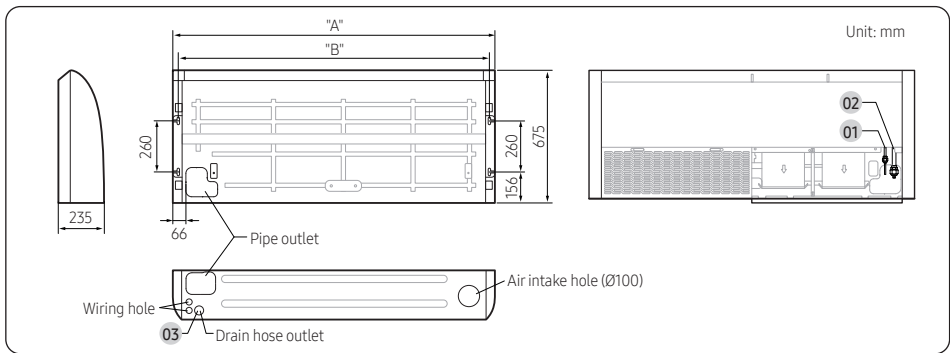
※ Floor Installation Not Available

Dimension of the indoor unit

Ceiling type (AM056DNC**G***)



Big Ceiling type (AM071/112/140DNC**G***)

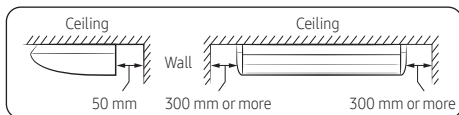


Model	Dimension		1	2	3
	A	B	Liquid pipe connection	Gas pipe connection	Drain pipe connection
Ceiling type (AM056DNC**G***)	1000	922	ø6.35 (1/4")	ø12.7 (1/2")	ID ø18 Hose
Big Ceiling type (AM071/112DNC**G***)	1350	1298	ø9.52 (3/8")	ø15.88 (5/8")	OD ø25; ID ø20
Big Ceiling type (AM140DNC**G***)	1650	1598	ø9.52 (3/8")	ø15.88 (5/8")	OD ø25; ID ø20

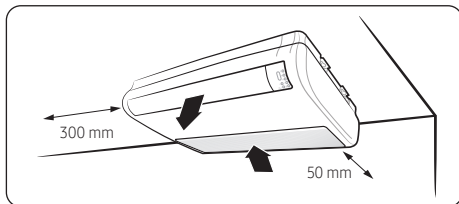
Installation Procedure

Spacing requirements

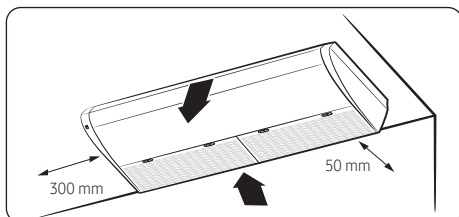
It is recommended to install the Y-joint before installing the indoor unit.



Ceiling type (AM056DNC**G***)



Big Ceiling type (AM071/112/140DNC**G***)



CAUTION

- Floor Installation Not Available.

Step 3 Installing the indoor unit

Ceiling installation

Ceiling type (AM056DNC**G***)

- Select pipe directions.

When the directions are selected, drill 3-1/8"-(100mm, for pipe and cables) and 1-3/4"-(40mm, for drain hose) diameter holes on the wall so that it slants slightly downwards toward the outdoor for smooth water flow.

NOTE

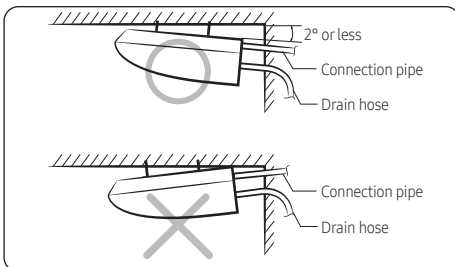
- Use the pattern sheet to select pipe directions.
- Drill holes for anchor bolts according to the distance and mount them.

NOTE

- Use the pattern sheet.
- Install the unit onto the ceiling. Be sure to arrange the drain hose so that it is leveled lower than the drain hose connecting port of the indoor unit.

NOTE

- For proper drainage of condensate, give a 2° or less slant to the side of the unit which will be connected with the drain hose as shown in the figure. (The gap between the lower end of the indoor unit and the ceiling should be 12 mm or less.)

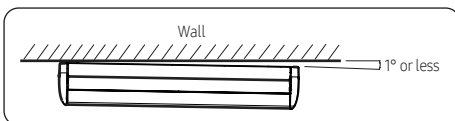


CAUTION

- Ensure that the ceiling is strong enough to support the weight of the indoor unit.
- Before hanging the unit, test the strength of each attached suspension bolt.
- Install the drain hose from the rear of the unit.

NOTE

- The gap between the lower end of the indoor unit and the ceiling should be 1° (16 mm) or less.



Ceiling installation

Big Ceiling type (AM071/112/140DNC**G***)

1 Select pipe directions.

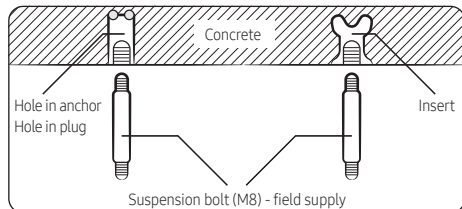
When the directions are selected, drill 3-1/8"-(100mm, for pipe and cables) and 1-3/4"-(40mm, for drain hose) diameter holes on the wall so that it slants slightly downwards toward the outdoor for smooth water flow.

NOTE

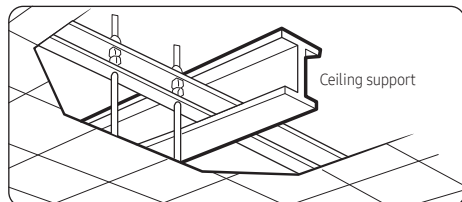
- Use the pattern sheet to select pipe directions.
- #### 2 Drill holes for anchor bolts according to the distance and mount them.

NOTE

- Use the pattern sheet.
- #### 3 Insert bolt anchors. Use existing ceiling supports or construct a suitable support as shown in figure.



4 Install the suspension bolts depending on the ceiling type.



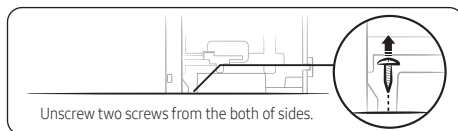
CAUTION

- Ensure that the ceiling is strong enough to support the weight of the indoor unit. Before hanging the unit, test the strength of each attached suspension bolt.
- If the length of suspension bolt is more than 1.5m, it is required to prevent vibration.
- If this is not possible, create an opening on the false ceiling in order to be able to use it to perform the required operations on the indoor unit.

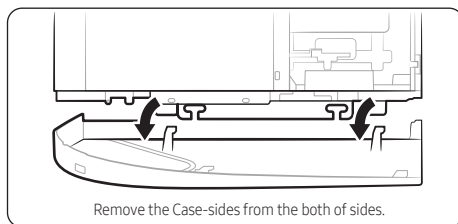
- #### 5 Screw eight nuts to the suspension bolts making space for hanging the indoor unit.

NOTE

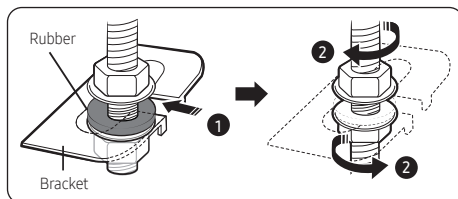
- You must install all the suspension rods.
- #### 6 When Hanging the set, firstly unscrew the screws from the both of sides, and then disassemble the Case-sides, or else the case-side will be damaged by disassembling it directly.



- #### 7 Reassemble the Case-sides, tightening the screws after hanging the set.



- #### 8 Hang the indoor unit to the suspension bolts between two nuts.



CAUTION

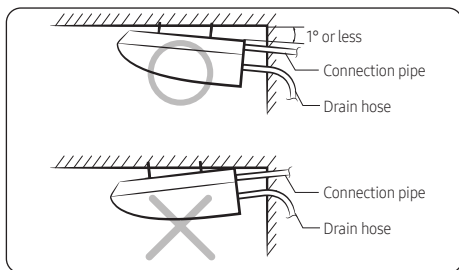
- Piping must be laid and connected inside the ceiling when suspending the unit. If the ceiling is already constructed, lay the piping into position for connection to the unit before placing the unit inside the ceiling.

Installation Procedure

- 9 Screw the nuts to suspend the unit.
- 10 Adjust level of the unit by using measurement plate for all 4 sides.

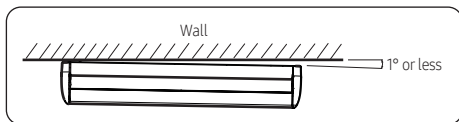
NOTE

- For proper drainage of condensate, give a 1° or less slant to the side of the unit which will be connected with the drain hose as shown in the figure. (The gap between the lower end of the indoor unit and the ceiling should be 12 mm or less.)



NOTE

- AM071/112DNC**G** : The gap between the lower end of the indoor unit and the ceiling should be 1° (23 mm) or less.
- AM140DNC**G** : The gap between the lower end of the indoor unit and the ceiling should be 1° (28 mm) or less.

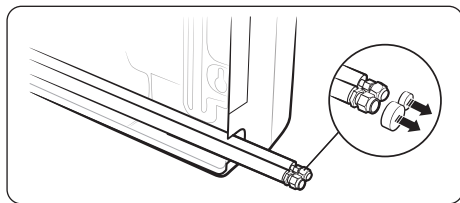


Step 4 Purging inert gas from the indoor unit

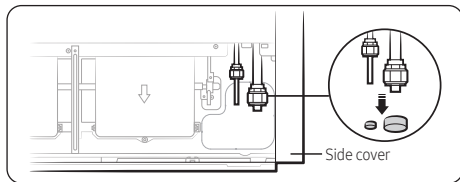
The indoor unit comes with nitrogen gas (inert gas) charged at the factory. Therefore, all inert gas must be purged before connecting the assembly piping.

Unscrew the pinch pipe at the end of each refrigerant pipe.

Ceiling type (AM056DNC**G***)



Big Ceiling type (AM071/112/140DNC**G***)

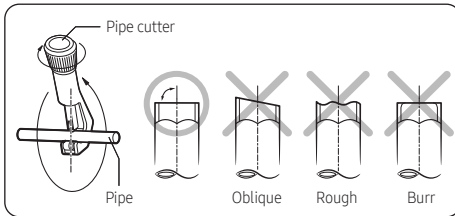


NOTE

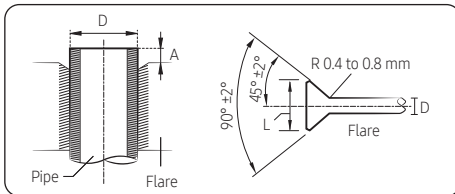
- To prevent dirt or foreign objects from getting into the pipes during installation, do not remove the pinch pipe completely until you are ready to connect the piping.
- The designs and shape are subject to change according to the model.

Step 5 Cutting and flaring the pipes

- Make sure that you have the required tools available: pipe cutter, reamer, flaring tool, and pipe holder.
- If you wish to shorten the pipes, cut them with a pipe cutter, ensuring that the cut edge remains at a 90° angle to the side of the pipe. Refer to the illustrations below for examples of edges cut correctly and incorrectly.

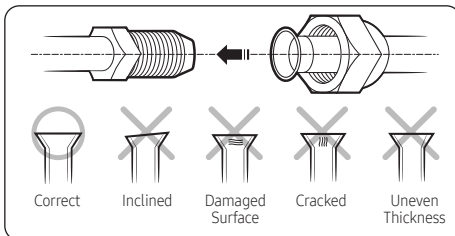


- 3 To prevent any gas from leaking out, remove all burrs at the cut edge of the pipe, using a reamer.
- 4 Slide a flare nut on to the pipe and modify the flare.



Outer Diameter (D)	Depth (A)	Flare dimension (L)
Ø6.35 mm	1.3 mm	8.7 to 9.1 mm
Ø9.52 mm	1.8 mm	12.8 to 13.2 mm
Ø12.70 mm	2.0 mm	16.2 to 16.6 mm
Ø15.88 mm	2.2 mm	19.3 to 19.7 mm
Ø19.05 mm	2.2 mm	23.6 to 24.0 mm

- 5 Check that the flaring is correct, referring to the illustrations below for examples of incorrect flaring.

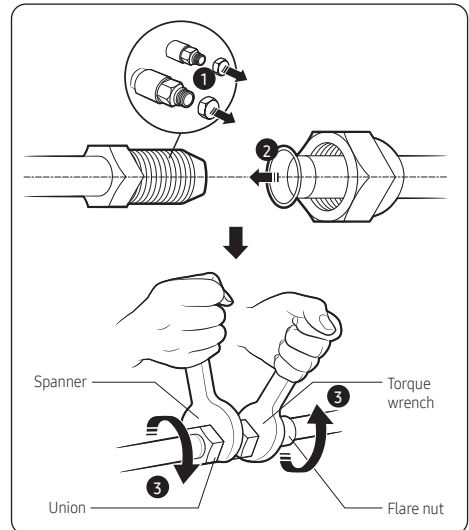


Step 6 Connecting the assembly pipes to the refrigerant pipes

There are two refrigerant pipes of different diameters :

- A smaller one for the liquid refrigerant.
- A larger one for the gas refrigerant. The inside of copper pipe must be clean and has no dust.

- 1 Remove the pinch pipe on the pipes and connect the assembly pipes to each pipe, tightening the nuts, first manually and then with a torque wrench, a spanner applying the following torque.



Outer diameter (mm)	Torque (N-m)
Ø6.35	14 to 18
Ø9.52	34 to 42
Ø12.70	49 to 61
Ø15.88	68 to 82
Ø19.05	100 to 120

(1 N-m=10 kgf-cm)

Installation Procedure

NOTE

- If the pipes must be shortened, see Step 5 Cutting and flaring the pipes on page 12.
 - Tighten the nuts to the specified torques. If overtightened, the nuts could be broken so refrigerant may leak.
- 2 Be sure to use an insulator thick enough to cover the refrigerant tube to protect the condensate water on the outside of the pipe falling onto the floor and to improve the efficiency of the unit.
 - 3 Cut off any excess foam insulation.
 - 4 Make sure that there are no cracks or waves on the bent area.
 - 5 It would be necessary to double the insulation thickness (10 mm or more) to prevent condensation even on the insulator when if the installed area is warm and humid.

CAUTION

- Connect the indoor and outdoor units using pipes with flared connections (not supplied). For the lines, use insulated, unwelded, degreased and deoxidized copper pipe (Cu DHP type to ISO 1337 or UNI EN 12735-1), suitable for operating pressures of at least 4.2 MPa and for a burst pressure of at least 20.7 MPa. Copper pipe for hydro-sanitary applications is completely unsuitable.
- For sizing and limits (height difference, line length, max. bends, refrigerant charge, etc.) see the outdoor unit installation manual.
- All refrigerant connection must be accessible, in order to permit either unit maintenance or removing it completely.
- If the pipes require brazing, make sure that oxygen free nitrogen (OFN) is flowing through the system.
- Nitrogen blowing pressure range is 0.02 to 0.05 MPa.

Step 7 Performing the gas leak test

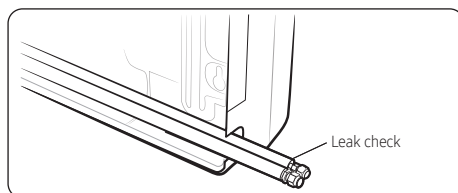
LEAK TEST WITH NITROGEN (before opening valves)

In order to detect basic refrigerant leaks, before recreating the vacuum and recirculating the R-32, it's responsible of installer to pressurize the whole system with nitrogen (using a cylinder with pressure reducer) at a pressure above 40 bar (gauge).

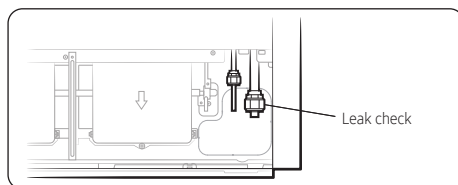
LEAK TEST WITH R-32 (after opening valves)

Before opening valves, discharge all the nitrogen into the system and create vacuum. After opening valves check leaks using a leak detector for refrigerant R-32.

Ceiling type (AM056DNC**G***)



Big Ceiling type (AM071/112/140DNC**G***)



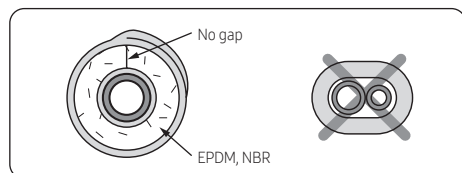
NOTE

- The designs and shape are subject to change according to the model.

Step 8 Insulating the refrigerant pipes

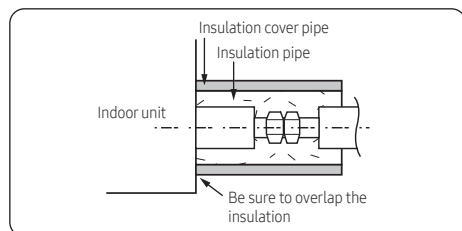
Once you have checked that there are no leaks in the system, you can insulate the piping and hose.

- 1 To avoid condensation problems, place Acrylonitrile Butadiene Rubber separately around each refrigerant pipe.



NOTE

- Always make the seam of pipes face upwards.
- 2 Wind insulating tape around the pipes and drain hose avoiding to compress the insulation too much.
- 3 Finish wrapping insulating tape around the rest of the pipes leading to the outdoor unit.
- 4 The pipes and electrical cables connecting the indoor unit with the outdoor unit must be fixed to the wall with suitable ducts.



CAUTION

- Must fit tightly against body without any gap.
- All refrigerant connection must be accessible, in order to permit either unit maintenance or removing it completely.

- 5 Select the insulation of the refrigerant pipe.
 - Insulate the gas side and liquid side pipe referring to the thickness according to the pipe size.
 - Indoor temperature of 30 °C and humidity of 85 % is the standard condition.
 - If installing in a high humidity condition, use one grade thicker insulator by referring to the table below.
 - If installing in an unfavorable conditions, use thicker one.
 - Insulator's heat-resistance temperature should be more than 120 °C.

Pipe	Pipe size	Insulation Type (Heating/Cooling)		Remarks
		Standard [30 °C, 85 %]	High humidity [30 °C, over 85 %]	
		EPDM, NBR		
Liquid pipe	Ø 6.35 ~ Ø 9.52	9 t	9 t	Internal temperature is higher than 120 °C
	Ø 12.7 ~ Ø 19.05	13 t	13 t	
Gas pipe	Ø 6.35	13 t	19 t	
	Ø 9.52	19 t	25 t	
	Ø 12.70			
	Ø 15.88			
	Ø 19.05			

When installing insulation in places and conditions below, use the same insulation that is used for high humidity conditions

<Geological condition>

- High humidity places such as shoreline, hot spring, near lake or river, and ridge (when the part of the building is covered by earth and sand.)

<Operation purpose condition>

- Restaurant ceiling, sauna, swimming pool etc.

<Building construction condition>

- The ceiling frequently exposed to moisture and cooling is not covered.

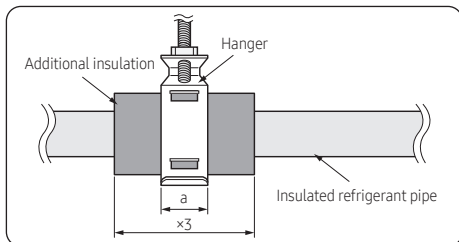
e.g. The pipe installed at a corridor of a dormitory and studio or near an exit that opens and closes frequently.

- The place where the pipe is installed is highly humid due to the lack of ventilation system.

Installation Procedure

⚠ CAUTION

- Install the insulation not to get wider and use the adhesives on the connection part of it to prevent moisture from entering.
- Wind the refrigerant pipe with insulation tape if it is exposed to outside sunlight.
- Install the refrigerant pipe respecting that the insulation does not get thinner on the bent part or hanger of pipe.
- Add the additional insulation if the insulation plate gets thinner.



Step 9 Installing the drain hose and drain pipe

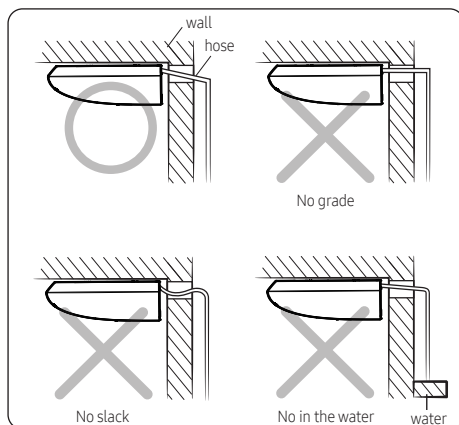
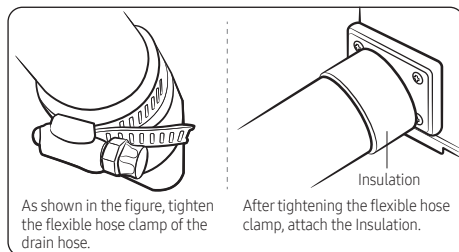
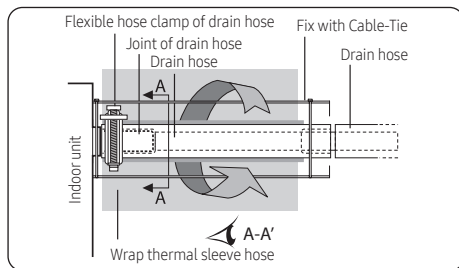
Ceiling type

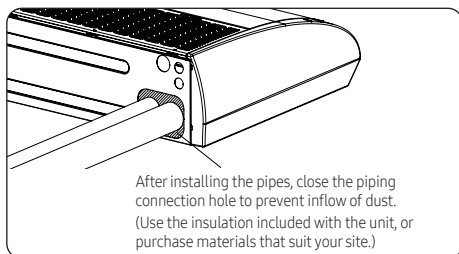
Care must be taken when installing the drain hose for the indoor unit to ensure that any condensate water is correctly drained outside. When passing the drain hose through the hole drilled in the wall, check that none of the following situations occur.

The drain hose can be installed to the right of the base pan.
(Big Ceiling type : AM071/112/140DNC**G***)

- 1 Installing the drain hose should be the shorter, the better.
 - In order to discharge condensation water, the drain hose should keep tilted.
 - Fix the drain hose with Cable-Tie, so that it will not separate from the machine.

- 2 Insulate and fix the drain hose according to the figure.
 - Insert the drain hose to bottom of the outfall of water basin.
 - Lock flexible hose clamp of the drain hose according to the figure.
 - Wind and wrap flexible hose clamp and drain hose fully with thermal insulation sponge; fix both ends of external layer with ribbon for thermal insulation.
 - After being installed, drain hose must be insulated fully by heat insulating material. (To be provided at site.)





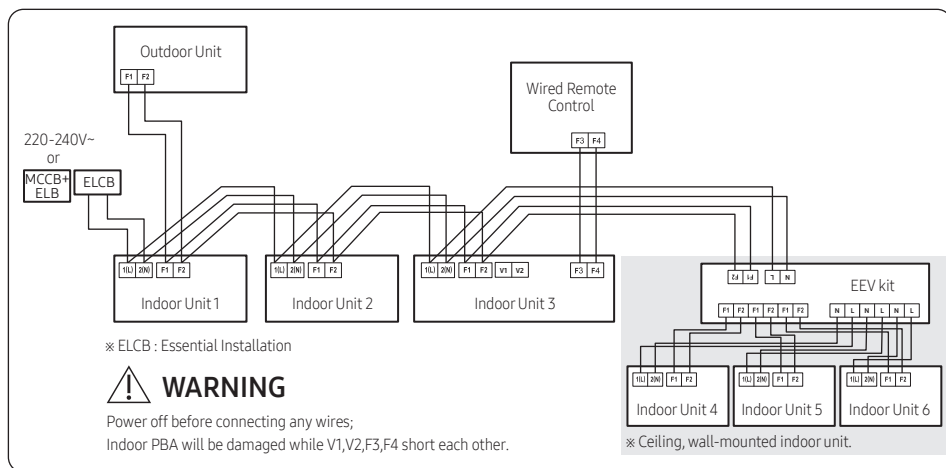
 NOTE

- Please follow national and local electrical codes. Additional electrical connection components may be required.

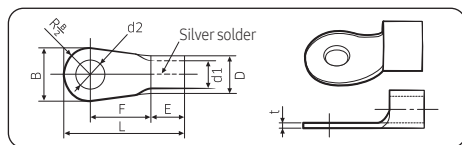
Installation Procedure

Step 10 Connecting the power and communication cables

- 1 Before wiring work, you must turn off all power source.
- 2 Indoor unit power should be supplied through the breaker(ELCB or MCCB+ELB) separated by the outdoor power.
 - ELCB : Earth Leakage Circuit Breaker
 - MCCB : Molded Case Circuit Breaker
 - ELB : Earth Leakage Breaker
- 3 The power cable should be used only copper wires.
- 4 Connect the power cable(1(L), 2(N)) among the units within maximum length and communication cable(F1, F2) each.
- 5 Connect F3, F4(for communication) when installing the wired remote control.



Selecting compressed ring terminal



Normal dimensions for cable (mm ²)	Normal dimensions for screw (mm)	B		D		d1		E	F	L	d2		t
		Standard dimension (mm)	Allowance (mm)	Standard dimension (mm)	Allowance (mm)	Standard dimension (mm)	Allowance (mm)	Min.	Min.	Max.	Standard dimension (mm)	Allowance (mm)	Min.
1.5	4	6.6	±0.2	3.4	+0.3 -0.2	1.7	±0.2	4.1	6	16	4.3	+0.2 0	0.7
	4	8											
2.5	4	6.6	±0.2	4.2	+0.3 -0.2	2.3	±0.2	6	6	17.5	4.3	+0.2 0	0.8
	4	8.5											
4	4	9.5	±0.2	5.6	+0.3 -0.2	3.4	±0.2	6	5	20	4.3	+0.2 0	0.9

Specification of electronic wire

Power supply	MCCB	ELB or ELCB	Power cable	Earth cable	Communication cable
Max : 242V Min : 198V	XA	XA, 30mmA 0.1 s	2.5mm ²	2.5mm ²	0.75-1.5mm ²

- Decide the capacity of ELCB(or MCCB+ELB) by below formula.
- Power supply cords of parts of appliances for outdoor use shall not be lighter than polychloroprene sheathed flexible cord.

(Code designation IEC:60245 IEC 57 / CENELEC: H05RN-F or IEC:60245 IEC 66 / CENELEC: H07RN-F)

The capacity of ELCB(or MCCB+ELB) $X[A] = 1.25 \times 1.1 \times \sum A_i$

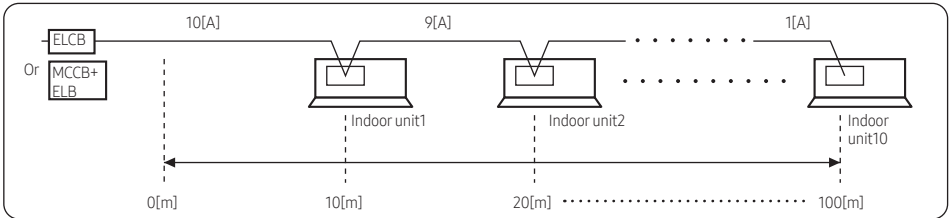
- X : The capacity of ELCB(or MCCB+ELB).
- $\sum A_i$: Sum of Rating currents of each indoor unit.
- Refer to each installation manual about the rating current of indoor unit.
- Decide the power cable specification and maximum length within 10% power drop among indoor units.

$$\sum_{k=1}^n \left(\frac{\text{Coef} \times 35.6 \times L_k \times i_k}{1000 \times A_k} \right) < 10\% \text{ of input voltage}[V]$$

- Coef: 1.55
- L_k : Distance among each indoor unit[m],
- A_k : Power cable specification[mm²]
- i_k : Running current of each unit[A]

Example of Installation

- Total power cable length $L = 100(m)$, Running current of each units 1[A]
- Total 10 indoor units were installed



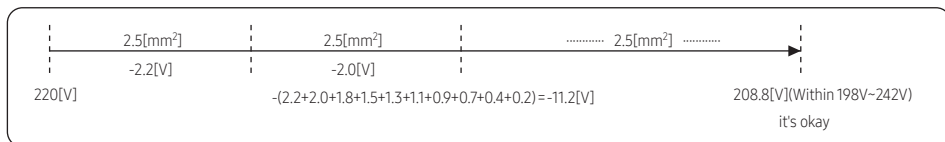
- Apply following equation.

$$\sum_{k=1}^n \left(\frac{\text{Coef} \times 35.6 \times L_k \times i_k}{1000 \times A_k} \right) < 10\% \text{ of input voltage}[V]$$

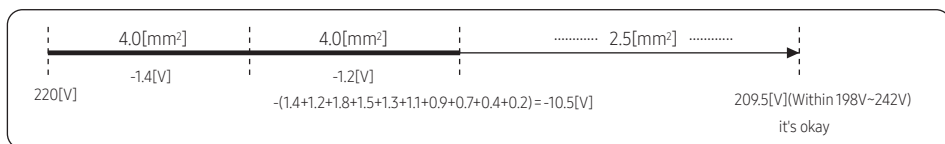
Installation Procedure

Calculation

- Installing with 1 sort wire.



- Installing with 2 different sort wire.



Rating current

Unit	Model	Rating current
Ceiling type	AM056DNC**G***	0.43A
Big Ceiling type	AM071DNC**G***	0.47A
	AM112DNC**G***	0.94A
	AM140DNC**G***	1.45A

CAUTION

- Select the power cable in accordance with relevant local and national regulations.
- Wire size must comply with local and national code.
- For the power cable, use the grade of H07RN-F or H05RN-F materials.
- You should connect the power cable into the power cable terminal and fasten it with a clamp.
- The unbalanced power must be maintained within 10% of supply rating among whole indoor units.
- If the power is unbalanced greatly, it may shorten the life of the condenser. If the unbalanced power is exceeded over 10% of supply rating, the indoor unit is protected, stopped and the error mode indicates.
- To protect the product from water and possible shock, you should keep the power cable and the connection cord of the indoor and outdoor units in the iron pipe.
- Connect the power cable to the auxiliary circuit breaker. An all pole disconnection from the power supply must be incorporated in the fixed wiring($\geq 3mm$).
- You must keep the cable in a protection tube.

- Keep distances of 50mm or more between power cable and communication cable.
- Maximum length of power cables are decided within 10% of power drop. If it exceeds, you must consider another power supplying method.
- The circuit breaker(ELCB or MCCB+ELB) should be considered more capacity if many indoor units are connected from one breaker.
- Use round pressure terminal for connections to the power terminal block.
- For wiring, use the designated power cable and connect it firmly, then secure to prevent outside pressure being exerted on the terminal board.
- Use an appropriate screwdriver for tightening the terminal screws. A screwdriver with a small head will strip the head and make proper tightening impossible.
- Over-tightening the terminal screws may break them.
- See the table below for tightening torque for the terminal screws.

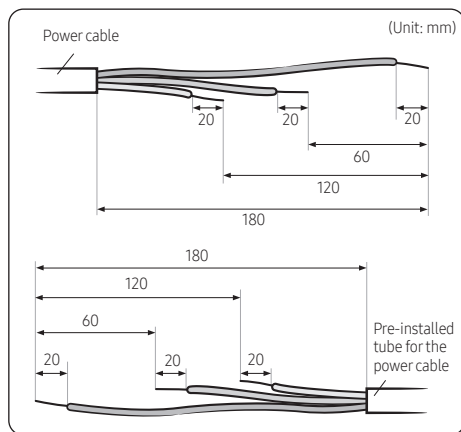
	Tightening torque	
	N•m	kgf•cm
M3.5	0.8~1.0	8.0~10.0
M4	1.2~1.5	12.0~14.7

Step 11 Optional: Extending the power cable

- 1 Prepare the following tools.

Tools	Spec	Shape
Crimping pliers	MH-14	
Connection sleeve (mm)	20 x Ø6.5 (H x OD)	
Insulation tape	Width 19 mm	
Contraction tube (mm)	70 x Ø8.0 (L x OD)	

- 2 As shown in the figure, peel off the shields from the rubber and wire of the power cable.
 - Peel off 20 mm of cable shields from the pre-installed tube.

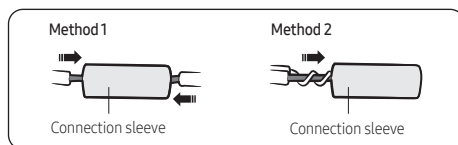


CAUTION

- For information about the power cable specifications for indoor and outdoor units, refer to the installation manual.
- After peeling off cable wires from the pre-installed tube, insert a contraction tube.

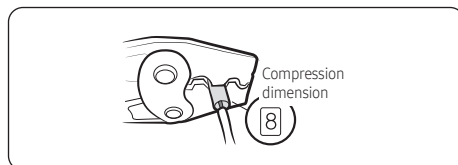
- 3 Insert both sides of core wire of the power cable into the connection sleeve.

- Method 1: Push the core wire into the sleeve from both sides.
- Method 2: Twist the wire cores together and push it into the sleeve.

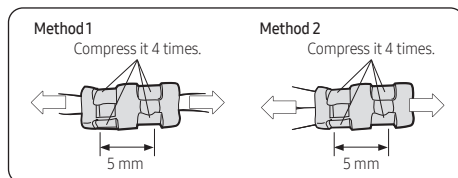


CAUTION

- If cable wires are connected without using connecting sleeves, their contact area becomes reduced, or corrosion develops on the outer surfaces of the wires (copper wires) over a long time. This may cause an increase of resistance (reduction of passing current) and consequently may result in a fire.
- 4 Using a crimping tool, compress the two points and flip it over and compress another two points in the same location.
 - The compression dimension should be 8.0.

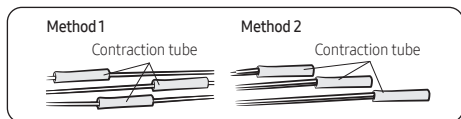


- After compressing it, pull both sides of the wire to make sure it is firmly pressed.

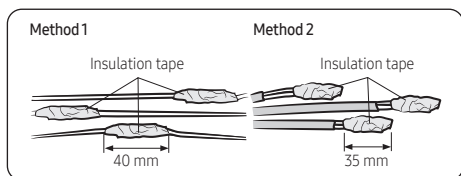


Installation Procedure

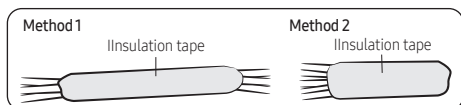
- 5 Apply heat to the contraction tube to contract it.



- 6 Wrap it with the insulation tape twice or more and position your contraction tube in the middle of the insulation tape.



- 7 After tube contraction work is completed, wrap it with the insulation tape to finish. Three or more layers of insulation are required.

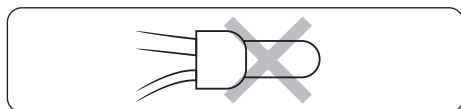


CAUTION

- Make sure that the connection parts are not exposed to outside.
- Be sure to use insulation tape and a contraction tube made of approved reinforced insulating materials that have the same level of withstand voltage with the power cable. (Comply with the local regulations on extensions.)

WARNING

- In case of extending the electric wire, please DO NOT use a round-shaped Pressing socket.
 - Incomplete wire connections can cause electric shock or a fire.

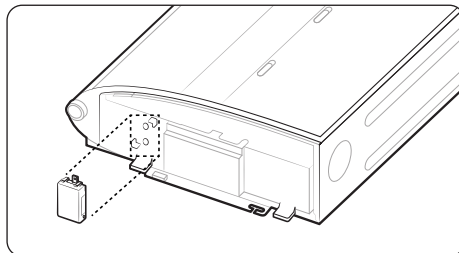


Step 12 Optional: Installing an interface module

Interface module : MIM-B14

External Control	PCB Case
Haness Wire(2P)	Haness Wire (4P)
Screw	

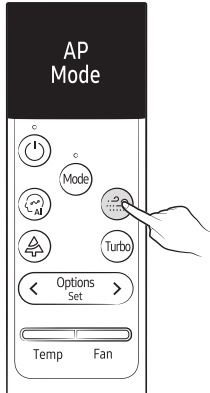
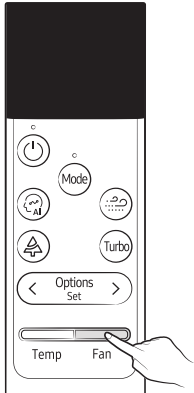
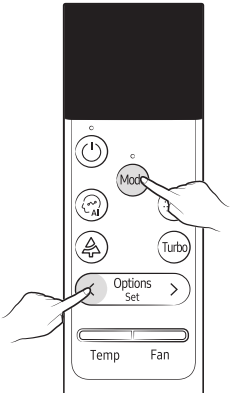
- 1 Fix the case at with bolts on the side of the control box in the indoor unit.(See the picture)



- 2 Attach the Interface module PCB to the case in the control box of the indoor unit, then connect the power and the communication cable between the Interface module and the indoor unit;
- 3 If you install a Interface module to an indoor unit, every outdoor unit which is connected to an indoor unit can be controlled simultaneously.
- 4 Each indoor unit connected to the same centralized controller has its own Interface module.

Step 13 Optional : LED Display indicator specifications when checking Wi-Fi Easy Setup and Wi-Fi status

The wireless remote control can be used for Easy Setup, checking the internet connection status and connecting or disconnecting Wi-Fi.

Easy Setup	Check internet connection status	Enable/Disable Wi-Fi
		
Press the  button for 5 seconds	Press the  button for 5 seconds	Press the  and  button for 5 seconds

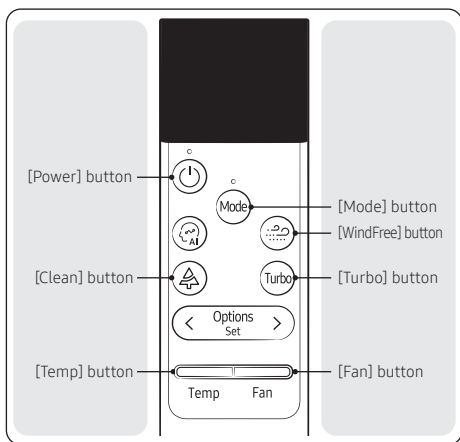
Installation Procedure

Step 14 Setting the indoor unit addresses and the installation options

You cannot set both indoor unit addresses and the installation options in a batch: set both respectively.

Common steps for setting the addresses and options

Remote controls

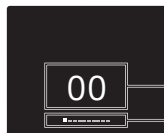


NOTE

- The remote control display and buttons may vary depending on the model.

1 Enter the mode for setting the options.

- Reset remote control: button Down + button Down + Press for 10 seconds
- You can see the "SW Initialization" message and enter the following in 5 seconds.
- Press button and button for 5 seconds.
- Make sure that you are entered to the mode for setting options.



2 Digit

10 point (2SEG per point)
ex) now SEG2 and SEG3

2 Set the option values.

CAUTION

- The total number of available options are 24 : **SEG1** to **SEG24**
- Because **SEG1**, **SEG7**, **SEG13** and **SEG19** are the page options used by the previous remote control models, the modes to set values for these options are skipped automatically.
- Set a 2-digit value for each option pair in the following order.
- You can see 20 **SEG** (except **SEG1**, **SEG7**, **SEG13**, **SEG19**)
SEG2 → ... → **SEG6** → **SEG8** → → **SEG12** → **SEG14** → ... → **SEG18** → **SEG20** → ... → **SEG24**

SEG1	SEG2	SEG3	SEG4	SEG5	SEG6
0	X	X	X	X	X
SEG7	SEG8	SEG9	SEG10	SEG11	SEG12
1	X	X	X	X	X
SEG13	SEG14	SEG15	SEG16	SEG17	SEG18
2	X	X	X	X	X
SEG19	SEG20	SEG21	SEG22	SEG23	SEG24
3	X	X	X	X	X

- You can set the next **SEG** by pressing the .
- You can change the digit value through the following operation.
Left value: up or down, range : 0 ~ F
Right value: up or down, range : 0 ~ F

Take the steps presented in the following table:

Steps	Remote control display
1 Set the SEG2 and SEG3 values: a Set the SEG2 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG3 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ... →  → 	 SEG2  SEG3
2 Press the  button to move to the next page.	
3 Set the SEG4 and SEG5 values: a Set the SEG4 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG5 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ... →  → 	 SEG4  SEG5
4 Press the  button to move to the next page.	

Installation Procedure

Steps	Remote control display
5 Set the SEG6 and SEG8 values: a Set the SEG6 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG8 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ... →  → 	 SEG6  SEG8
6 Press the  button to move to the next page.	
7 Set the SEG9 and SEG10 values: a Set the SEG9 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG10 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ... →  → 	 SEG9  SEG10
8 Press the  button to move to the next page.	

Steps	Remote control display
9 Set the SEG11 and SEG12 values: a Set the SEG11 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG12 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ... →  → 	 SEG11  SEG12
10 Press the  button to move to the next page.	
11 Set the SEG14 and SEG15 values: a Set the SEG14 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG15 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ... →  → 	 SEG14  SEG15
12 Press the  button to move to the next page.	

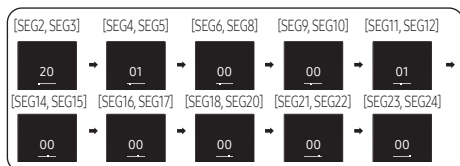
Installation Procedure

Steps	Remote control display
13 Set the SEG16 and SEG17 values: a Set the SEG16 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG17 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order: 0 → 8 → ... E → F	 SEG16  SEG17
14 Press the  button to move to the next page.	
15 Set the SEG18 and SEG20 values: a Set the SEG18 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG20 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order: 0 → 8 → ... E → F	 SEG18  SEG20
16 Press the  button to move to the next page.	

Steps	Remote control display
17 Set the SEG21 and SEG22 values: a Set the SEG21 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG22 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ...  → 	 SEG21  SEG22
18 Press the  button to move to the next page.	
19 Set the SEG23 and SEG24 values: a Set the SEG23 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. b Set the SEG24 value by pressing the  button repeatedly until the value you want to set appears on the remote control display. When you press the  or  button, values appear in the following order:  →  → ...  → 	 SEG23  SEG24

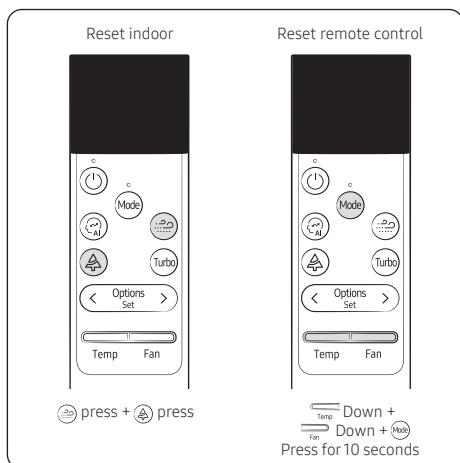
Installation Procedure

- 3 Check whether the option values you have set are correct by pressing the **Mode** button repeatedly.



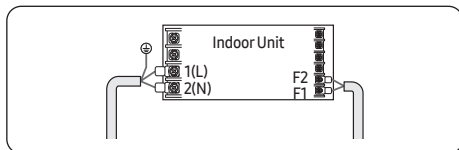
EX) **AM***DNCDKG**
020010-100000-2000A0-300000

- 4 Save the option values into the indoor unit:
Point the remote control to the remote control sensor on the indoor unit and then press the **Power** button on the remote control twice.
Make sure that this command is received by the indoor unit. When it is successfully received, you can hear a short sound from the indoor unit. If the command is not received, press the **Power** button again.
- 5 Check whether the air conditioner operates following the option values you have set:
- Reset the indoor or outdoor unit.
 - Indoor Unit : Press **Power** button + **Mode** button for 5 seconds
 - Outdoor Unit : Press the **K3** button
 - Reset remote control: **Temp** button Down + **Fan** button Down + **Mode** Press for 10 seconds
You can see the "**SW Initialization**" message.



Setting an indoor unit address (MAIN/RMC)

- 1 Check whether power is supplied or not.
 - When the indoor unit is not plugged in, there should be additional power supply in the indoor unit.
- 2 The panel(display) should be connected to an indoor unit to receive option.
- 3 Before installing the indoor unit, assign an address to the indoor unit according to the air conditioning system plan.
- 4 Assign an indoor unit address by wireless remote controller.
 - The initial setting status of indoor unit ADDRESS(MAIN/RMC) is "0A0000-100000-200000-300000".



Option No. : 0AXXXX-1XXXXX-2XXXXX-3XXXXX

Option	SEG1		SEG2		SEG3		SEG4		SEG5		SEG6	
Explanation	PAGE		Mode		Setting Main address		100-digit of indoor unit address		10-digit of indoor unit		The unit digit of an indoor unit	
Indication and Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details
	0		A		0	No Main address	0~9	100-digit	0~9	10-digit	0~9	A unit digit
					1	Main address setting mode						
Option	SEG7		SEG8		SEG9		SEG10		SEG11		SEG12	
Explanation	PAGE				Setting RMC address				Group channel(x16)		Group address	
Indication and Details	Indication	Details			Indication	Details			Indication	Details	Indication	Details
	1		-		0	No RMC address	-		RMC1	0~F	RMC2	0~F
					1	RMC address setting mode						



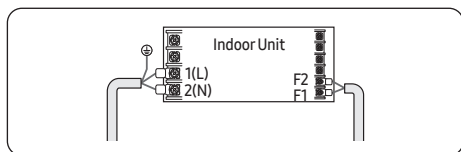
CAUTION

- When "A"~"F" is entered to SEG5~6, the indoor unit MAIN ADDRESS is not changed.
- If you set the SEG 3 as 0, the indoor unit will maintain the previous MAIN ADDRESS even if you input the option value of SEG5~6.
- If you set the SEG 9 as 0, the indoor unit will maintain previous RMC ADDRESS even if you input the option value of SEG11~12.
- You cannot set SEG11 and SEG12 as F value at the same time.

Installation Procedure

Setting an indoor unit installation option (suitable for the condition of each installation location)

- 1 Check whether power is supplied or not.
 - When the indoor unit is not plugged in, there should be additional power supply in the indoor unit.
- 2 The panel(display) should be connected to an indoor unit to receive option.
- 3 Set the installation option according to the installation condition of an air conditioner.
 - The default setting of an indoor unit installation option is "020010-100000- 2000A0-300000".
 - Individual control of a remote controller(SEG20) is the function that controls an indoor unit individually when there is more than one indoor unit.
- 4 Set the indoor unit option by wireless remote controller.



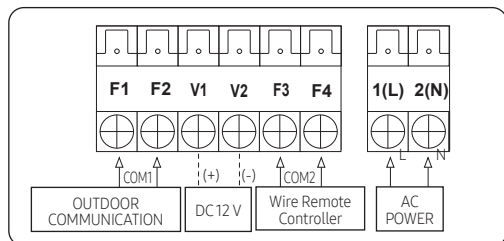
Option No. : 0AXXXX-1XXXXX-2XXXXX-3XXXXX

02 series installation option

SEG1	SEG2	SEG3	SEG4	SEG5	SEG6
0	2	Evaporator Drying	Use of external room temperature sensor / Minimizing fan operation when thermostat is off	Use of central control	FAN RPM compensation
SEG7	SEG8	SEG9	SEG10	SEG11	SEG12
1	Use of drain pump	Use of hot water heater	Settings for load operation during heater control Fan control during defrost mode / heater control during defrost mode	EEV Step when heating stops	Dew removal operation in wind free mode
SEG13	SEG14	SEG15	SEG16	SEG17	SEG18
2	Use of external control	Setting the output of external control / External heater signal / Cooling operation signal / Free Cooling control signal	S-Plasma ion	Buzzer control / whether to use humidity sensor / whether to use APP UX DSP (Dual Set Point) / whether to use R-32 sensor	Hours of filter usage
SEG19	SEG20	SEG21	SEG22	SEG23	SEG24
3	Individual control of a remote controller	Heating setting compensation / Removing condensate water in heating mode	Adjusted EEV step of stopped unit during oil return /defrost mode.	Motion detect sensor	-

- 1WAY/2WAY/4WAY MODEL : Drain pump(SEG8) will be set to 'USE + 3minute delay' even if the drain pump is set to 0.
- 1 WAY/2WAY/4WAY,DUCT MODEL : Number of hours using filter(SEG18) will be set to '1000hour' even if the SEG18 is set to except for 2 or 6.
- When setting the option other than above SEG values, the option will be set as "0".
- SEG5 central control option is basically set as 1 (Use), so you don't need to set the central control option additionally. However, if the central control is not connected but it doesn't indicate an error message, you need to set the central control option as 0 (Disuse) to exclude the indoor unit from the central control.

- The output of hot water heater in SEG9 is generated from the hot coil part of the terminal board in duct models.



- * The output of hot coil terminal is AC 220 V / 230 V
(The same as Indoor Unit's input Power)

- The external output of SEG15 is generated by MIM-B14 connection. (Refer to the manual of MIM-B14.)

02 series installation option(Detailed)

Option No. : 02XXXX-1XXXXX-2XXXXX-3XXXXX

Option	SEG1		SEG2		SEG3		SEG4			SEG5		SEG6	
Explanation	PAGE		MODE		Evaporator Drying		Use of external room temperature sensor / Minimizing fan operation when thermostat is off			Use of central control		FAN RPM compensation	
Indication and Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details		Indication	Details	Indication	Details
	0		2	0	Disuse	0		Default	Minimizing fan operation when thermostat is off				
						1	Use	Disuse					
						2	Disuse	Use (Heating) (*2)					
					2	Use (5min) (*1)	3	Use	Use (Heating) (*2)	1	RPM compensation		
								4	Disuse			Use (Cooling) (*2)	
								5	Use			Use (Cooling) (*2)	
					4	Use (10min) (*1)	6	Disuse	Use (Heating / Cooling) (*2)	1	Use		
								7	Use			Use (Heating / Cooling) (*2)	
								8	Disuse			Use (Cooling Ultra Low Fan) (*2)	
					6	Use (30min) (*1)	9	Use	Use (Cooling Ultra Low Fan) (*2)	2	High ceiling KIT (4way model only)		
								A	Disuse			Use (Heating / Cooling Ultra Low Fan) (*2)	
								B	Use			Use (Heating / Cooling Ultra Low Fan) (*2)	

Installation Procedure

Option	SEG7		SEG8		SEG9		SEG10			SEG11		SEG12	
Explanation	PAGE		Use of drain pump		Use of hot water heater		Settings for load operation during heater control Fan control during defrost mode / Heater control during defrost mode			EEV Step when heating stops		Dew removal operation in wind free mode	
Indication and Details	Indication	Details	Indication	Details	Indication	Details	Indication	Detail		Indication	Details	Indication	Details
	1		0	Disuse	0	Disuse	0	Fan Off	Off	0	Default	0	(Default) Maintain blade status in wind free mode
							1	Fan turns on when heater turns on	Off				
							2	Fan Off	Off				
							3	Fan turns on when heater turns on	Off				
							4	Fan Off	On				
							5	Fan turns on when heater turns on	On				
			1	Use	1	Use (*3)	6	Fan Off	On	1	Adjusted EEV Step setting	1	Cooling operation by opening the blade
							7	Fan turns on when heater turns on	On				
							8	Fan Off	Off				
							9	Fan turns on when heater turns on	Off				
							A	Fan Off	Off				
							B	Fan turns on when heater turns on	Off				
							C	Fan Off	On				
			2	When an indoor unit stops, drain pump will operate for 3min	3	Use (*3)	D	Fan turns on when heater turns on	On				
							E	Fan Off	On				
							F	Fan turns on when heater turns on	On				

Option	SEG13		SEG14		SEG15		SEG16		SEG17					SEG18			
Explanation	PAGE		Use of external control		Setting the output of external control / External heater signal / Cooling operation signal / Free Cooling control signal		S-Plasma ion		Buzzer control / whether to use humidity sensor / whether to use APP UX DSP (Dual Set Point) / whether to use R-32 sensor					Hours of filter usage			
Indication and Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Detail				Indication	Details		
	2		0	Disuse	0	External control (Thermo On)	0	Disuse	0	Use Buzzer	Disuse	Disuse	Disuse	2	1000 Hour		
									1	Disuse Buzzer	Disuse	Disuse	Disuse				
			1	ON/OFF control	1	External control (Operation On)	2	Use Buzzer	Use	Disuse	Disuse	2	1000 Hour				
							3	Disuse Buzzer	Use	Disuse	Disuse						
							4	Use Buzzer	Disuse	Use	Disuse						
			2	OFF control	3	External heater signal (*4)	5	Disuse Buzzer	Disuse	Use	Disuse	2	1000 Hour				
							6	Use Buzzer	Use	Use	Disuse						
							7	Disuse Buzzer	Use	Use	Disuse						
			3	Window ON/OFF control	4	Cooling operation signal (*5)	5	Free Cooling control (Cooling Thermo On) (*6)	1	Use	8	Use Buzzer	Disuse	Disuse	Use	6	2000 Hour
											9	Disuse Buzzer	Disuse	Disuse	Use		
											A	Use Buzzer	Use	Disuse	Use		
					B	Disuse Buzzer	Use	Disuse	Use								
					C	Use Buzzer	Disuse	Use	Use								
					D	Disuse Buzzer	Disuse	Use	Use								
E	Use Buzzer	Use	Use	Use													
F	Disuse Buzzer	Use	Use	Use													

Installation Procedure

Option	SEG19		SEG20		SEG21			SEG22		SEG23		SEG24	
Explanation	PAGE		Individual control of a remote controller		Heating setting compensation / Removing condensate water in heating mode			Adjusted EEV step of stopped unit during oil return /defrost mode.		Setting the MDS Kit installation option		-	
Indication and Details	Indication	Details	Indication	Details	Indication	Details		Indication	Details	Indication	Details	-	
						Heating Setting Compensation	Removing Condensate Water in Heating Mode						
	3		0 or 1	channel 1	0	Default	Disuse	0	Default	0	Disuse (Soft Off+Hard off) (*8)		
					1	2 °C	Disuse			1	Off after 20 min. (Soft Off+Hard off)		
			2	channel 2	2	5 °C	Disuse			2	Off after 40 min. (Soft Off+Hard off)		
					3	Default	Use (*7)			3	Off after 80 min. (Soft Off+Hard off)		
										4	Off after 20 min. (Soft Off+Hard off)		
			3	channel 3	4	2 °C	Use (*7)	1	Adjusted EEV position	5	Off after 40 min. (Soft Off+Hard off)		
										6	Off after 80 min. (Soft Off+Hard off)		
			4	channel 4						7	Off after 20 min. (Soft Off only) (*9)		
										8	Off after 40 min. (Soft Off only)		
										9	Off after 80 min. (Soft Off only)		
			5							A	Off after 20 min. (Soft Off only)		
										B	Off after 40 min. (Soft Off only)		
C										Off after 80 min. (Soft Off only)			

* Advanced function: Controlling cooling/heating current or power saving with motion detect.

(*1) When Cooling or dry mode is off. The indoor fan operate in setting minutes.

(*2) Minimizing fan operation when thermostat is off

- Fan operates for 20 seconds at an interval of 5 minutes in Heat mode.

- Fan stops or operates Ultra low in Cooling when thermostat is off.

(*3) 1: Fan is turned on continually when the hot water heater is turned on,

3: Fan is turned off when the hot water heater is turned on with cooling only indoor unit

Cooling only indoor unit: To use this option, install the Mode Select switch(MCM-C200) on the outdoor unit and fix it as Cool mode.

(*4) When the following 2 or 3 is used as external heater On/Off signal, the signal for monitoring external contact control will not be output.

2: Fan is turned on continually when the external heater is turned on,

3: Fan is turned off when the external heater is turned on with cooling only indoor unit

Cooling only indoor unit: To use this option, install the Mode Select switch(MCM-C200) on the outdoor unit and fix it as Cool mode.

- If Fan is set to off for cooling only indoor unit by setting the SEG9=3 or SEG15=3, you need to use an external sensor or wired remote controller sensor to detect indoor temperature exactly.

(*5) When indoor unit is in cooling or Dry mode, The output signal is "ON"

(*6) For free cooling control, Economizer controller is required.

(*7) This function can be applied to 4 Way Cassette and Mini 4 Way Cassette only.

If the air conditioner operates the heating mode immediately after finishing the cooling mode, the condensate water in the drain pan becomes water vapor by the heat of the indoor unit heat exchanger. Since the water vapor might be condensed on the indoor unit, which may fall into a living space, use this function to get rid of the water vapor out of the indoor unit by operating the fan (for maximum 20 minutes) even when the indoor unit is turned off after cooling mode is turned to heating mode.

(*8) Soft Off: If no motion is detected for the Soft Off time, the MDS Kit turns off the indoor units.

Then if any motion is detected until the Hard Off time is passed, the MDS Kit restarts the indoor units.

(*9) Hard Off: If no motion is detected for the Hard Off time, the MDS Kit turns hard off the indoor units.

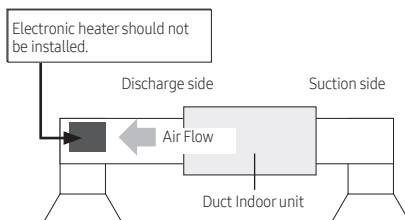
Then although any motion is detected, the MDS Kit does not restart the indoor and outdoor units.

You must manually restart the units with the wired or wireless remote control.



CAUTION

- Do not install the electronic heater in the flow channel of the indoor unit fan.



Installation Procedure

05 series installation option

SEG1	SEG2	SEG3	SEG4	SEG5	SEG6
0	5	Use of Auto Change Over for HR only in Auto mode / Use of Cooling only indoor unit of HR	(When setting SEG3) Standard heating temp. Offset	(When setting SEG3) Standard cooling temp. Offset	(When setting SEG3) Standard for mode change Heating → Cooling
SEG7	SEG8	SEG9	SEG10	SEG11	SEG12
1	(When setting SEG3) Standard for mode change Cooling → Heating	(When setting SEG3) Time required for mode change	Compensation option for Long pipe or height difference between indoor units	MTFC (*3)	-
SEG13	SEG14	SEG15	SEG16	SEG17	SEG18
2	-	Dual fuel (heater lock) setting	Dual fuel (HP lock) setting	-	Control variables when using hot water / external heater (*4)
SEG19	SEG20	SEG21	SEG22	SEG23	SEG24
3	-	-	-	Forced FAN Operation for Heating and Cooling	Whether to use UV LED / whether to use BLE Onboarding / whether to allow fan speed control during auto mode / MDS (motion detection sensor) control UX type

05 series installation option(Detailed)

Option No. : 05XXXX-1XXXXX-2XXXXX-3XXXXX

Option	SEG1		SEG2		SEG3		SEG4		SEG5		SEG6	
Explanation	PAGE		MODE		Use of Auto Change Over for HR only in Auto mode / Use of Cooling only indoor unit of HR		(When setting SEG3) Standard heating temp. Offset		(When setting SEG3) Standard cooling temp. Offset		(When setting SEG3) Standard for mode change Heating → Cooling	
Indication and Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details
	0		5		0	Follow product option	0	0°C	0	0°C	0	1°C
							1	0.5°C	1	0.5°C	1	1.5°C
					1	Use Auto Change Over for HR only	2	1°C	2	1°C	2	2°C
							3	1.5°C	3	1.5°C	3	2.5°C
							4	2°C	4	2°C	4	3°C
					2	Use Cooling only indoor unit for HR	5	2.5°C	5	2.5°C	5	3.5°C
							6	3°C	6	3°C	6	4°C
							7	3.5°C	7	3.5°C	7	4.5°C

Option	SEG7		SEG8		SEG9		SEG10		SEG11		SEG12	
Explanation	PAGE		(When setting SEG3) Standard for mode change Cooling → Heating		(When setting SEG3) Time required for mode change		Compensation option for Long pipe or height difference between indoor units		MTFC (*3)		-	
Indication and Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details		
	1		0	1°C	0	5 min.	0	Default	0	Default		
			1	1.5°C	1	7 min.	1	(*1) Height difference is more than 30m or (*2) Distance is longer than 110m				
			2	2°C	2	9 min.						
			3	2.5°C	3	11 min.						
			4	3°C	4	13 min.	2	(*1) Height difference is 15~30m or (*2) Distance is 50~110m	2	Use		
			5	3.5°C	5	15 min.						
			6	4°C	6	20 min.						
			7	4.5°C	7	30 min.						

Option	SEG13		SEG14	SEG15		SEG16		SEG17	SEG18		
Explanation	-		-	Dual fuel (heater lock) setting		Dual fuel (HP lock) setting		-	Control variables when using hot water / external heater (*4)		
Indication and Details	Indication	Details	-	Indication	Detail	Indication	Detail	-	Indication	Details	
	2		-					-		0	Disuse
				0	Disuse	0	Disuse		0	At the same time as thermo on	No delay
				1	18.3 °C	1	72 °C		1	At the same time as thermo on	10 minutes
				2	15.6 °C	2	4.4 °C		2	At the same time as thermo on	20 minutes
				3	12.8 °C	3	17 °C		3	1.5 °C	No delay
				4	10 °C	4	-11 °C		4	1.5 °C	10 minutes
				5	72 °C	5	-3.9 °C		5	1.5 °C	20 minutes
				6	4.4 °C	6	-6.7 °C		6	3.0 °C	No delay
				7	17 °C	7	-9.4 °C		7	3.0 °C	10 minutes
				8	-11 °C	8	-12.2 °C		8	3.0 °C	20 minutes
				9	-3.9 °C	9	-15 °C		9	4.5 °C	No delay
				A	-6.7 °C	A	-17.8 °C		A	4.5 °C	10 minutes
				B	-9.4 °C	B	-20.6 °C		B	4.5 °C	20 minutes
				C	-12.2 °C	C	-23 °C		C	6.0 °C	No delay
				D	-15 °C	D	-26 °C		D	6.0 °C	10 minutes
				E	-17.8 °C	E	-29 °C		E	6.0 °C	20 minutes
				F	Cannot be used	F	Cannot be used				

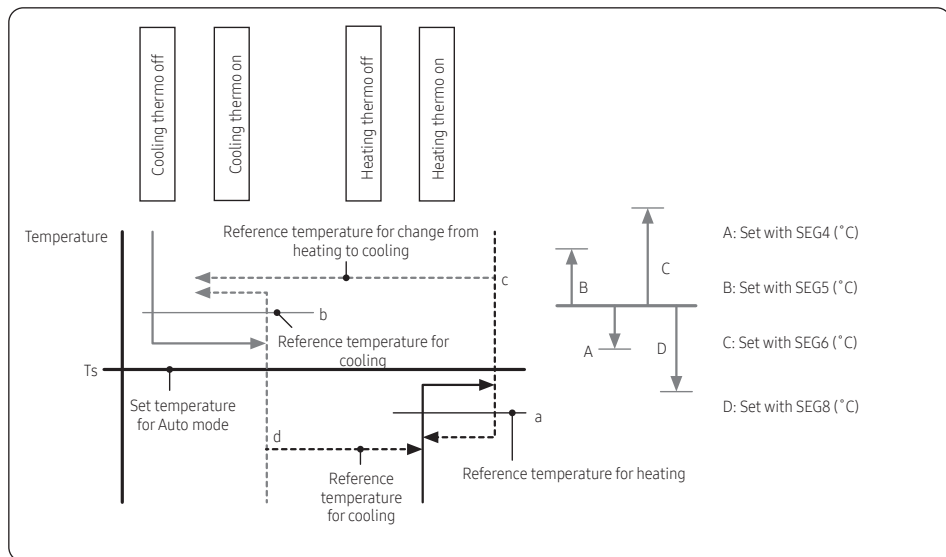
Installation Procedure

Option	SEG19		SEG20	SEG21	SEG22	SEG23		SEG24				
Explanation	PAGE		-	-	-	Forcing FAN Operation for Heating and Cooling		Whether to use UV LED / whether to use BLE Onboarding / whether to allow fan speed control during auto mode / MDS (motion detection sensor) control UX type				
	Indication	Details				Indication	Details		Indication	Detail		
							Cooling Fan Setting	Heating Fan Setting		UV LED	BLE Onboarding	MDS (motion detection sensor) control UX type
Indication and Details	3		-	-	-	0	Disuse	Disuse	0	Disuse	Disuse	Fan speed and power saving mode can be set simultaneously
						1	Disuse	Use (Fan: User setting)	1	Use	Disuse	Fan speed and power saving mode can be set simultaneously
						2	Disuse	Use (Fan: High)	2	Disuse	Use	Fan speed and power saving mode can be set simultaneously
						3	Disuse	Use (Fan: Low)	3	Use	Use	Fan speed and power saving mode can be set simultaneously
						4	Use (Fan: User setting)	Disuse	4	Disuse	Disuse	Fan speed and power saving mode can be set simultaneously
						5	Use (Fan: User setting)	Use (Fan: User setting)	5	Use	Disuse	Fan speed and power saving mode can be set simultaneously
						6	Use (Fan: User setting)	Use (Fan: High)	6	Disuse	Use	Fan speed and power saving mode can be set simultaneously
						7	Use (Fan: User setting)	Use (Fan: Low)	7	Use	Use	Fan speed and power saving mode can be set simultaneously
						8	Use (Fan: High)	Disuse	8	Disuse	Disuse	Only fan speed or power saving mode can be set at a time
						9	Use (Fan: High)	Use (Fan: User setting)	9	Use	Disuse	Only fan speed or power saving mode can be set at a time
						A	Use (Fan: High)	Use (Fan: High)	A	Disuse	Use	Only fan speed or power saving mode can be set at a time
						B	Use (Fan: High)	Use (Fan: Low)	B	Use	Use	Only fan speed or power saving mode can be set at a time
						C	Use (Fan: Low)	Disuse	C	Disuse	Disuse	Only fan speed or power saving mode can be set at a time
						D	Use (Fan: Low)	Use (Fan: User setting)	D	Use	Disuse	Only fan speed or power saving mode can be set at a time
						E	Use (Fan: Low)	Use (Fan: High)	E	Disuse	Use	Only fan speed or power saving mode can be set at a time
						F	Use (Fan: Low)	Use (Fan: Low)	F	Use	Use	Only fan speed or power saving mode can be set at a time

- (*1) Height difference : The difference of the height between the corresponding indoor unit and the indoor unit installed at the lowest place. For example, When the indoor unit is installed 40m higher than the indoor unit installed at the lowest place, select the option "1".
- (*2) The difference between the pipe length of the indoor unit installed at farthest place from an outdoor unit and the pipe length of the corresponding indoor unit from an outdoor unit.
For example, when the farthest pipe length is 100 m(328 ft.) and the corresponding indoor unit is 40 m away from an outdoor unit, select the option "2". (100 - 40 = 60m)
- (*3) For MTFC option, MTFC(Multi Tenant Function Controller) kit is required.
- (*4) Heater operation when the SEG9 of O2 series installation option is set to using hot water heater or when SEG15 is set to using external heater.
Example 1) Setting O2 series SEG9 = "1" / Setting O5 series SEG18 = "0": The hot water heater is turned on at the same time as the heating thermostat is on, and turned off when the heating thermostat is off.
Example 2) Setting O2 series SEG15 = "2" / Setting O5 series SEG18 = "A": Room temp. $\leq$ set temp. + f (heating compensation temp.)
- External heater is turned on when the temperature is maintained as 4.5 °C for 10 minutes. Room temp. $>$ set temp. + f(heating compensation temp.)
- External heater is turned off when the temperature is maintained as 4.5 °C + 1 °C (1 °C is the Hysteresis for On/Off selection.)

Additional information on SEG3, 4, 5, 6, 8, 9

When SEG3 is set to 1 and the HR-specific auto changeover function is run, the indoor unit operates as shown in the following figure:



The mode change between the Cool and Heat modes is made only when the thermo off state is maintained for the period of time set with SEG9.

Installation Procedure

Changing a particular option

You can change each digit of set option.

Option	SEG1		SEG2		SEG3		SEG4		SEG5		SEG6	
Explanation	PAGE		MODE		Type of the option to change		Tens position of the option number		Units position of the option number		New value	
Indication and Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details	Indication	Details
	0		D		Option mode	0~F	Tens position value	0~9	Units position value	0~9	New value	0~F

NOTE

- When changing a digit of an indoor unit address setting option, set the SEG3 as 'A'.
 - When changing a digit of indoor unit installation option, set the SEG3 as '2'.
- Ex) When setting the 'buzzer control' into disuse status.

Option	SEG1	SEG2	SEG3	SEG4	SEG5	SEG6
Explanation	PAGE	MODE	Type of the option to change	Tens position of the option number	Units position of the option number	New value
Indication	0	D	2	1	7	1

CAUTION

- If your indoor units support both cooling and heating, the mixed operation (two or more indoor units operate in different modes simultaneously) is not available when the indoor units are connected to the same outdoor unit. If you set an indoor unit as the master indoor unit by using the remote control, the outdoor unit automatically operate in the current mode of the master indoor unit.

Installing external outputs

An external output signal occurs if the R-32 sensor in the indoor unit detects a refrigerant leak, or the sensor has a malfunction or short circuit.

Based on this signal, safety measures required for the indoor unit, such as ventilation system activation and alarm activation, can be taken.

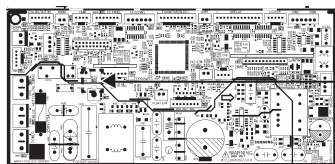
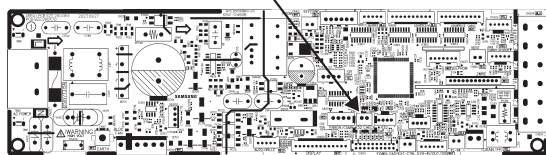
MIM-B14** (External Contact Control Module) can be used to link the GAS LEAK output.

NOTE

- The MIM-B14** can be connected to the required load on connectors 3 and 4.
- The load is AC (220~240), AC 2.25Amax
- When an error occurs due to a gas leak or R-32 sensor error, 3 and 4 are in a short state (the relay operates).

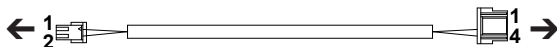
For controlling AC LAMP (On/Off)

Ceiling : R-32 CHECK : CN802(YELLOW)

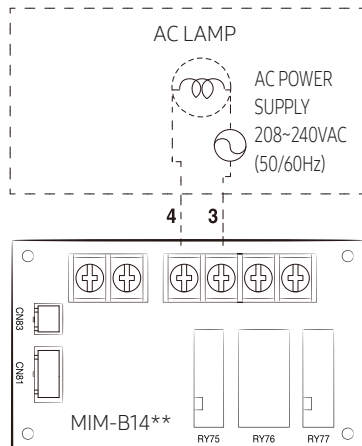


Big Ceiling : R-32
CHECK : CN802(BLUE)

※ Use of the WIRE HARNESS included in the product manual

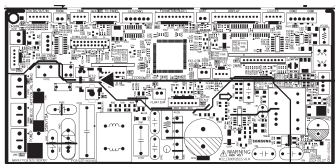
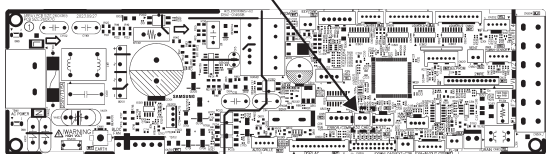


To Main PBA



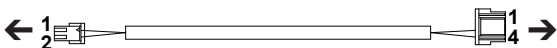
For controlling EXTERNAL ALARM (On/Off)

Ceiling : R-32 CHECK : CN802(YELLOW)

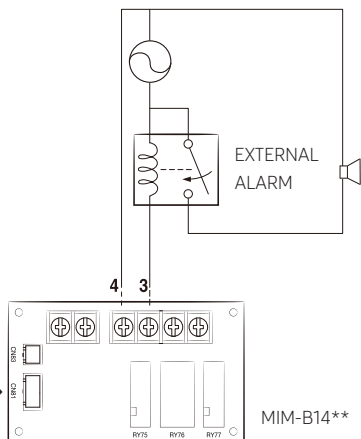


Big Ceiling : R-32
CHECK : CN802(BLUE)

※ Use of the WIRE HARNESS included in the product manual



To Main PBA



Troubleshooting

Ceiling type (AM056DNC**G***)

- If an error occurs during the operation, one or more LED flickers and the operation is stopped except the LED.
 - If you re-operate the air conditioner, it operates normally at first, then detect an error again.
- On ● Flickering X Off
- If you turn off the air conditioner when the LED is flickering, the LED is also turned off.
 - If you re-operate the air conditioner, it operates normally at first, then detect an error again.
 - When E108 error occurs, change the address and reset the system.Ex.) When address of the indoor unit #1 and #2 are set as 5, address of the indoor unit #1 will become 5 and indoor unit #2 will display E108, A002.

LED Display on the receiver & display unit

Abnormal condition	Error code	LED Display				
						
Error on indoor temperature sensor (Short or Open)	E121	X	●	X	X	X
1. Error on Eva-in sensor (Short or Open)	E122	●	●			
2. Error on Eva-out sensor (Short or Open)	E123	●	●	X	X	X
3. Discharge sensor error (Short or Open)	E126					
Indoor fan error	E154	X	X	●	X	X
1. Error on outdoor temperature sensor (Short or Open)	E221					
2. Error on cond sensor	E237	●	X	●	X	X
3. Error on discharge sensor	E251					
Other outdoor unit sensor error that is not on the above list						
1. When there is no communication between the indoor-outdoor units for 2 minutes	E101					
2. Communication error received from the outdoor unit	E102					
3. 3 minute tracking error on outdoor unit	E202					
4. Communication error after tracking due to unmatching number of installed units	E201	X	●	●	X	X
5. Error due to repeated communication address	E108					
6. Communication address not confirmed	E109					
Other outdoor unit communication error that is not on the above list						
Self diagnosis error display						
1. Error due to opened EEV (2nd detection)	E151					
2. Error due to closed EEV (2nd detection)	E152					
3. Eva in sensor is detached	E128	X	●	●	●	X
4. Eva out sensor is detached	E1289					
5. Thermal fuse error (Open)	E198					

Abnormal condition	Error code	LED Display				
						
1. COND mid sensor is detached	E241					
2. Refrigerant leakage (2nd detection)	E554					
3. Abnormally high temperature on Cond (2nd detection)	E450					
4. Low pressure s/w (2nd detection)	E451					
5. Abnormally high temperature on discharged air on outdoor unit(2nd detection)	E416					
6. Indoor operation stop due to unconfirmed error on outdoor unit	E559					
7. Error due to reverse phase detection	E425					
8. Comp stop due to freeze detection (6th detection)	E403					
9. High pressure sensor is detached	E301					
10. Low pressure sensor is detached	E306					
11. Outdoor unit copression ration error	E428					
12. Outdoor sump down_1 prevetion control	E413					
13. Compressor down due to low pressure sensor prevention control_1	E410	X				X
14. Simultaneous opening of cooling/heating MCU SOL valve (1st detection)	E180					
15. Simultaneous opening of cooling/heating MCU SOL valve (2nd detection)	E181					
16. Indoor unit R-32 sensor short/open	E116					
17. Installation combination of indoor unit and wired remote control Error	E694					
18. Refrigerant leak sensor lifetime unpredictable error	E695					
19. 1st refrigerant leak detection error	E696					
20. 2nd refrigerant leak detection error (Error-causing indoor unit)	E697					
21. Refrigerant leak sensor failure error	E698					
22. Refrigerant leak sensor replacement notification error	E699					
23. Refrigerant leak sensor lifetime expiration error	E700					
24. 2nd refrigerant leak detection error (Not Error-causing indoor unit)	E797					
Other outdoor unit self-diagnosis error that is not on the above list						
Flowating s/w (2nd detection)	E153	X	X			X
EEPROM error	E162					
EEPROM option error	E163					
Error due to incompatible indoor unit	E164	X	X	X		X

Troubleshooting

Big Ceiling type (AM071/112/140DNC**G***)

- If an error occurs during the operation, one or more LED flickers and the operation is stopped except the LED.
- If you re-operate the air conditioner, it operates normally at first, then detect an error again.

● On ○ Flickering X Off

- If you turn off the air conditioner when the LED is flickering, the LED is also turned off.
- If you re-operate the air conditioner, it operates normally at first, then detects an error again.
- If the LED displays only one color, it is turned on for a second and turned off for a second.
- If the LED displays more than two colors, each color is shown for a second alternately.

LED Display on the indoor unit

Abnormal conditions	Indicators				Remarks
	Blue	Green	Orange	Red	
Power reset	○	X	X	X	0.5[S]=On, 0.5[S]=Off
Operation on	●	X	X	X	
Operation off	X	X	X	X	
Reservation	X	●	X	X	
Filter sign	X	X	●	X	
Defrosting	○	X	X	X	1[S]=On, 9[S]=Off
Communication error between indoor units	X	○	X	X	
EEPROM error /EEPROM option error	○	X	X	○	
Error of temperature sensor in indoor unit (open/short)	X	X	X	○	
Error of outdoor Unit/Self-Diagnosis	X	X	○	X	
Error of the indoor unit pipe sensor	X	○	X	○	
High pressure blockage error	X	X	○	X	
Indoor fan error	○	○	X	X	
Thermal fuse open error	○	X	○	X	
Indoor unit float S/W 2nd detection	X	○	○	X	

Wired remote control

- If an error occurs,  is displayed on the wired remote control.
- If you would like to see an error code, press the Test button.

Display	Explanation
E108	Error due to repeated communication address
E116	Indoor unit R-32 sensor short/open
E121	Error on room temperature sensor of indoor unit (Short or Open)
E122	Error on EVA IN sensor of indoor unit (Short or Open)
E123	Error on EVA OUT sensor of indoor unit (Short or Open)
E153	Error on float switch (2nd detection)
E154	Indoor fan error
E198	Error on thermal fuse of indoor unit (Open)
E201	Communication error between indoor unit and outdoor unit (Pre tracking failure or when actual number of indoor units are different from the indoor unit quantity setting on the outdoor unit) Error due to communication tracking failure after initial power is supplied.
E202	Communication error between indoor unit and outdoor unit (When there is no response from indoor units after tracking is completed)
E203	Communication error between outdoor unit inv - main micom (For PF #4~#6 controller, error will be determined from the time when compressor is turned on)
E221	Error on outdoor temperature sensor (Short or Open)
E231	Error on outdoor COND OUT sensor (Short or Open)
E251	Error on discharge temperature sensor of compressor1 (Short or Open)
E320	Error on OLP sensor (Short or Open)
E403	Compressor down due to freeze protection control
E404	System stop due to overload protection control
E416	System stop due to discharge temperature
E422	Blockage detected on high pressure pipe
E425	Reverse phase or open phase
E440	Heating operation restricted at outdoor temperature over Theat_high value (default: 30 °C)
E441	Cooling operation restricted at outdoor temperature below Tcool_low value (default: 0 °C)
E458	Fan speed error
E461	Error due to operation failure of inverter compressor
E462	System stop due to full current control
E463	Over current trip / PFC over current error

Troubleshooting

Display	Explanation
E464	IPM Over Current(O.C)
E465	Comp. Over load error
E466	DC-Link voltage under/over error
E467	Error due to abnormal rotation of the compressor or unconnected wire of compressor
E468	Error on current sensor (Short or Open)
E469	Error on DC-Link voltage sensor (Short or Open)
E470	Outdoor unit EEPROM Read/Write error (Option)
E471	Outdoor unit EEPROM Read/Write error (H/W)
E472	AC Line Zero Cross Signal out
E473	Comp Lock error
E474	Error on IPM Heat Sink sensor of inverter1 (Short or Open)
E475	Error on inverter fan 2
E484	PFC Overload (Over current) Error
E485	Error on input current sensor of inverter1 (Short or Open)
E500	IPM over heat error on inverter1
E508	Smart install is not installed
E554	Gas leak detected
E556	Error due to mismatching capacity of indoor and outdoor unit
E557	DPM remote controller option error
E590	Inverter EEPROM CheckSum error
E660	Inverter Boot Code error
E694	Installation combination of indoor unit and wired remote control Error
E695	Refrigerant leak sensor lifetime unpredictable error
E696	1st refrigerant leak detection error
E697	2nd refrigerant leak detection error (Error-causing indoor unit)
E698	Refrigerant leak sensor failure error
E699	Refrigerant leak sensor replacement notification error
E700	Refrigerant leak sensor lifetime expiration error
E797	2nd refrigerant leak detection error (Not Error-causing indoor unit)



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